

Taiga Morishita

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Education:

Purdue University - West Lafayette

Graduating May 2028

B.S. Mechanical Engineering

- 3.85/4.00 GPA, Dean's List, 4x Semester Honors
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Experience:

Purdue Baja Racing

2024 - Current

Data Acquisition (DAQ) Lead

- Designing a wireless torsional strain measurement device for axle load testing
- Designing PCBs using **Altium Designer** for STM32-based fully custom DAQ system for 1st time in team history, **reducing total mass of DAQ by 37%** (2.5 lbs) in a single year
- Creating waterproof enclosures in **SOLIDWORKS** with custom-molded silicone seals for reliable & easy-to-use DAQ
- Installed a MTS wheel force transducer to measure 6-axis forces & moments exceeding 1 ton at 1 kSPS to measure sensor-synchronized loads across car and improve upcoming car design & analysis
- Configured a **LabJack-T7** to measure torsional strain and angular velocity for brand new custom-built combination engine-CVT dyno
- Wrote **MATLAB** scripts to interface with, process, and visualize various sources of data at up to 20kSPS
- 1st Place Overall at Baja SAE Oregon 2026 and 1st Place Design at Baja SAE New York 2026

Subaru of Indiana Automotive

Summer 2025

Body Manufacturing Engineering Intern

- Redesigned a Keyence laser profiler system to measure and report roof channels - exceeded expectations, successfully expanded changes to both production lines
- Programmed multiple **Mitsubishi Q-Series PLCs** to communicate with data collection system
- Updated ladder logic on over 100 Nachi robots (of 800+ across team) to simplify dry runs
- Debugged a doggedly failing camera system, adding failure recovery to permanently fix issue and prevent spurious quality assurance warnings

FIRST Robotics Competition Team 948 - Newport Robotics Group

2020 - 2024

VP Electrical

- Manage other students in a highly competitive, high-stress environment
 - Quality wiring of systems with momentary spikes of 200+A and hard impacts
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Projects:

Maze Solver Robot (Class Project)

March 2026

- Programmed a differential drive robot with a microcontroller to use a PI controller and odometry
 - Integrated custom rotary encoders into undercarriage to allow for closed-loop control
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Skills:

- CISCO Certified Network Associate (2023)
- Amazon Web Services Certified Cloud Practitioner (2024)
- MATLAB, SOLIDWORKS, Altium Designer, KiCad, NX, Onshape
- Fluent in Japanese